Midterm - Representations of Quivers

April 13th, 2026

Choose up to five problems from this list to submit.

1. Let Q be the quiver $1 \longleftarrow 2 \longrightarrow 3$ and let M be a representation

$$k \xleftarrow{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}} k^3 \xrightarrow{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}} k^2 .$$

   Write M as a direct sum of simple representations, indecomposable projective and injective representations.

2. Let Q be the quiver

$$\begin{tikzcd} 1 & 2 \arrow[l, shift right, "\alpha"'] \arrow[l, shift left, "\beta"] \end{tikzcd}$$

   For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$\begin{tikzcd} 1 & 2 \arrow[l, shift right, "1"'] \arrow[l, shift left, "\lambda"] \end{tikzcd} \qquad \lambda \in k;$$

$$\begin{tikzcd} 1 & 2 \arrow[l, shift right, "1"'] \arrow[l, shift left, "0"] \end{tikzcd} \qquad \lambda = \infty.$$

   (a) Show that each M_λ is indecomposable.
   (b) Show that $M_\lambda \simeq M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k.
   (c) Show that $\mathrm{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.

3. Let Q be a quiver, $L \xrightarrow{f} M \xrightarrow{g} N$ be an exact sequence in $rep\, Q$ and X be representation of Q.
   Show that $\mathrm{Ker}\, g_* \subseteq \mathrm{Im}\, f_*$, where f_* and g_* are defined in

$$\mathrm{Hom}(X, L) \xrightarrow{f_*} \mathrm{Hom}(X, M) \xrightarrow{g_*} \mathrm{Hom}(X, N) .$$

4. Let $g : L \to M$ be an injective morphism between representations of Q, and let $I(i)$ be the injective representation at vertex i. Show that the map

$$g^* : \mathrm{Hom}(M, I(i)) \to \mathrm{Hom}(L, I(i))$$

   is surjective.

5. Prove that any projective representation corresponding to vertex i, $P(i)$ is indecomposable.

6. Compute a projective resolutions for representations $S(1)$ and $S(2)$ for quiver

$$\begin{tikzcd} Q: & 1 \arrow[r] \arrow[rr, bend left] & 2 \arrow[r] \arrow[d] & 3 \arrow[dl] \\ & & 4 & \end{tikzcd}$$

7. Let $M = (M_i, \psi_\alpha)$ be a representation of Q. Prove that for any vertex i in Q, there is an isomorphism of vector spaces:

$$\mathrm{Hom}(M, I(i)) \simeq M_i.$$

8. Let i and j be vertices in the quiver Q. Prove that the vector space $\mathrm{Hom}(I(i), I(j))$ has a basis consisting of all paths from j to i in Q. In particular, $\mathrm{End}(I(i)) \simeq k$.